

**Indian Institute of Technology Jodhpur**  
**Department of Physics**  
**Mid-Sem Exam (PHL 1010)**

Duration: 2 Hours

18/09/2025

Max. Marks: 35

**Instructions:**

- Attempt all questions.
- Question 1 carries “completing the sentence” and finding “True/False” statements. Marks would be given only when you give a to-the-point justification/reasoning for your answer.
- Use pen only. Pencil can be used only for drawing figures.

1. Complete the following: [12]
- In 3-D cylindrical polar coordinate system, position vector  $\vec{r}$  has \_\_\_\_\_ nonzero components.
  - Length of the curve for which  $r=1$ ,  $\theta=30^\circ$ ,  $0 < \phi < 60^\circ$  is \_\_\_\_\_
  - For  $\vec{p}$  being a constant vector,  $\vec{\nabla} \times (\vec{p} \times \vec{r}) =$  \_\_\_\_\_
  - Dot product of any two general vector defined in spherical polar CS at points  $(1, \pi/2, 0)$  and  $(1, \pi/2, \pi/2)$  in  $\theta = \pi/2$  plane, will always be \_\_\_\_\_
  - For  $V$  being the volume of sphere of radius  $R$  with center at the origin,  
 $\int_V (r^3 + 5) \nabla^2 (1/r) d\tau =$  \_\_\_\_\_

Determine whether each of the following statements is True or False:

- $\vec{E} = yz \hat{x} + xz \hat{y} + xy \hat{z}$  is a possible electric field.
  - Electric field line shown in Fig. 1 in the region below the surface will bend away from the normal in the region above the boundary.
  - An irrotational field will always be uniquely expressed as a curl of potential.
  - Image of a dipole kept in front of a grounded conducting sphere will always be a dipole.
  - A point charge  $q$  is placed at a distance  $d$  from the center of an isolated neutral conducting sphere of radius  $a$  ( $a < d$ ). Potential on the surface of the sphere would be  $kq/d$ .
  - In a region, potential is defined by  $V = x^2z - yz$ . Potential as well as electric field both vanish at  $(1, -1, 0)$ .
  - Two spherical cavities are hollowed out from the interior of a neutral conducting sphere. Point charge  $q$  is placed in both the cavities at A and B. Another identical charge  $q$  is brought to point C in such a way that  $q$  at A is equidistant to  $q$  at B and  $q$  at C (Figure 2). Force on  $q$  at A will be  $45^\circ$  north to west.
2. Verify divergence theorem for a vector field defined by  $\vec{V} = r(2 + \sin^2 \phi) \hat{r} + r \sin \phi \cos \phi \hat{\phi} + 3z \hat{z}$  over the volume bounded by an annular cylinder of height  $h$ , inner radius  $R_1$  and outer radius  $R_2$ . The axis of the cylinder is along  $z$  – axis and one face is over  $z = 0$  plane (Figure 3). [5]

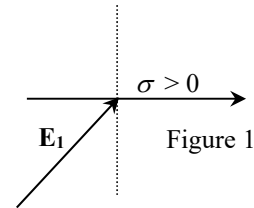


Figure 1

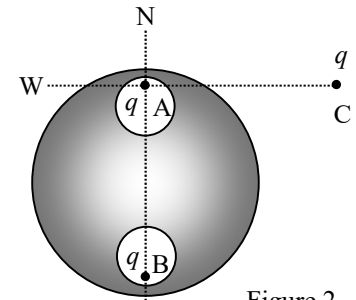


Figure 2

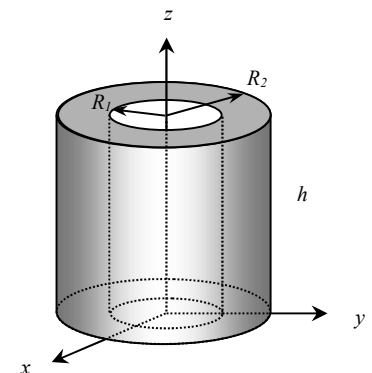


Figure 3

3. (a) Evaluate  $\int_V |\vec{r} - \vec{b}|^2 \delta^3(\vec{r} - \vec{a}) d\tau$  where  $\vec{b} = 4\hat{y} + 3\hat{z}$ ,  $\vec{a} = 4\hat{x} + 2\hat{y} + 3\hat{z}$  &  $V$  is a sphere of radius 5 centered at the origin.
- (b) A sphere of radius  $r$  carries a surface charge density  $\sigma = \vec{a} \cdot \vec{r}$ , where  $\vec{a}$  is a constant vector (you may assume it along  $+x$  direction) and  $\vec{r}$  is the radius vector of a point on the sphere w. r. t. its center. Show the charge distribution and find the electric field at the center of the sphere. [2+5]
4. (a) A point charge  $+q$  is kept outside at a distance  $d$  from the center of a grounded conducting sphere of radius  $a$ . Find the image charge and its location.
- (b) Consider the situation where a point charge  $+q$  is located inside a grounded conducting shell of radius  $a$ , halfway between the center and the shell (Fig. 4). Using the result obtained in (a), calculate
- Potential  $[V(x,y,z)]$  inside the sphere.
  - $\vec{E}$  inside the sphere. From this, evaluate  $\vec{E}(0,0,0)$  and  $\vec{E}(2a,0,0)$ .
  - Use  $\vec{E}(0,0,0)$  and  $\vec{E}(2a,0,0)$  to find  $\sigma(0,0,0)$  and  $\sigma(2a,0,0)$ .
  - What is the field outside the sphere? Can you get it with the method of images?

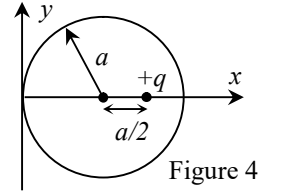


Figure 4

[5+6]

$$\vec{\nabla} \equiv \hat{\xi}_1 \frac{1}{h_1} \frac{\partial}{\partial \xi_1} + \hat{\xi}_2 \frac{1}{h_2} \frac{\partial}{\partial \xi_2} + \hat{\xi}_3 \frac{1}{h_3} \frac{\partial}{\partial \xi_3}$$

$$\vec{\nabla} \cdot \vec{V} = \frac{1}{h_1 h_2 h_3} \left[ \frac{\partial(V_1 h_2 h_3)}{\partial \xi_1} + \frac{\partial(V_2 h_3 h_1)}{\partial \xi_2} + \frac{\partial(V_3 h_1 h_2)}{\partial \xi_3} \right]$$

$$\vec{\nabla} \times \vec{V} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{\xi}_1 & h_2 \hat{\xi}_2 & h_3 \hat{\xi}_3 \\ \partial/\partial \xi_1 & \partial/\partial \xi_2 & \partial/\partial \xi_3 \\ h_1 V_1 & h_2 V_2 & h_3 V_3 \end{vmatrix}$$